

Matrix functions and stability

Realizing arbitrary polynomial growth exponents by finite matrix families

Difficulty: challenging **Importance:** interesting to the community

Status: open; explicit question in a 2025 journal paper

Context and notation

The joint spectral radius of a nonempty compact set $\mathcal{M} \subset \mathbb{C}^{d \times d}$ is

$$\hat{\rho}(\mathcal{M}) = \lim_{k \rightarrow \infty} \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

For a compact nonempty $\mathcal{M} \subset \mathbb{R}^{d \times d}$ with $\hat{\rho}(\mathcal{M}) = 1$, define its maximal product norm at length k by

$$g_{\mathcal{M}}(k) = \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2.$$

This problem concerns growth within one family over time. [MF-07](#) instead requests a dimension-dependent bound uniform across families.

Problem statement

For every real $\alpha \geq 0$, do there exist a positive integer d , a finite nonempty $\mathcal{M} \subset \mathbb{R}^{d \times d}$ with $\hat{\rho}(\mathcal{M}) = 1$, and constants $0 < c \leq C < \infty$ such that

$$ck^\alpha \leq g_{\mathcal{M}}(k) \leq Ck^\alpha \quad \text{for every integer } k \geq 1?$$

Reference and status evidence

Varney and Morris, [On marginal growth rates of matrix products](#), §7, Question 2. Corollary 6.1 realizes exponent $1/3$; the all-exponents question remains posed. Searches for the title, authors, and marginal-growth exponents through 2026 located no complete answer. Infinite compact families and subsequence-only lower bounds do not meet the finite-family, every-length target.